Customer Shopping Behavior Analysis

1. Project Overview

This project studies how customers shop in a retail store.
The dataset contains information about customer details and their purchases.

The goal of this project is to understand how customers buy products and what affects their buying decisions.

The analysis helps businesses improve marketing, increase sales, and understand customer behavior.

2. Dataset Summary

The dataset contains **3,900 rows** and **18 columns**.

The data includes:

Customer Information

- Age
- Gender
- Location
- Subscription status

Purchase Information

- Item purchased
- Category
- Purchase amount
- Season
- Size
- Color

Shopping Behavior

- Discount applied
- Previous purchases
- Frequency of purchases
- Review rating

- Shipping type

Some values were missing in the **review_rating** column and were handled during data cleaning.

3. Data Cleaning using Python

Python was used to prepare and clean the dataset.

Steps performed:

- Loaded the dataset using **Pandas**
- Checked the dataset structure
- Reviewed summary statistics
- Checked for missing values
- Filled missing values in the review rating column
- Renamed columns to make them easier to read
- Created a new column called **age_group**
- Converted purchase frequency into number of days

After cleaning the data, it was loaded into **MySQL** for further analysis.

4. Data Analysis using SQL

SQL queries were used to answer important business questions.

The analysis included:

Q1 — What is the total revenue generated by male vs female customers?

gender	total_revenue
Male	157890
Female	75191

Q2 — Which customers used a discount but still spent more than the average purchase amount?

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88

Q3 — Which are the top 5 products with the highest average review rating?

Result Grid Filter Rows:

item_purchased	avg_product_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

Q4 — Compare the average purchase amount between Standard and Express shipping.

Result Grid Filter Rows:

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

Q5 — Do subscribed customers spend more than non-subscribed customers?

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

Q6 — Which 5 products have the highest percentage of purchases with discounts?

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Q7 — Segment customers into New, Returning, and Loyal customers.

customer_segment	number_of_customers
Loyal	3116
Returning	701
New	83

Q8 — What are the top 3 most purchased products within each category?

item_rank	category	item_purchased	total_orders
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Q9 — Are repeat buyers more likely to subscribe?

subscription_status	repeat_buyers
Yes	958
No	2518

Q10 — What is the revenue contribution of each age group?

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

Q11 — Which locations generate the highest revenue?

location	total_revenue
Montana	5784
Illinois	5617
California	5605
Idaho	5587
Nevada	5514
Alabama	5261
New York	5257
North Dakota	5220
West Virginia	5174
Nebraska	5172

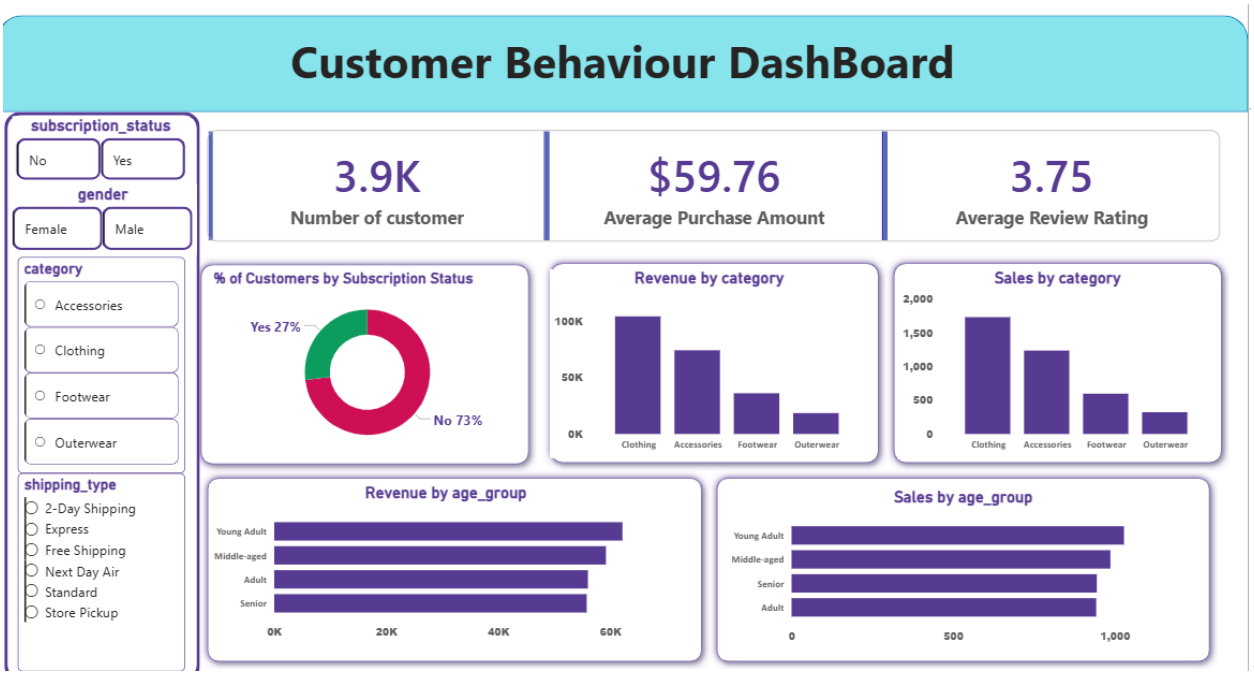
Q12 — Which product category generates the most revenue?

category	total_revenue	total_orders
Clothing	104264	1737
Accessories	74200	1240
Footwear	36093	599
Outerwear	18524	324

These analyses help understand customer shopping patterns.

5. Dashboard in Power BI

A Power BI dashboard was created to show the results of the analysis.



The dashboard shows key numbers such as:

- **Total Customers:** 3.9K
- **Average Purchase Amount:** $59.76
- **Average Review Rating:** 3.75

The dashboard also includes charts such as:

- Customer subscription status
- Revenue by category
- Sales by category
- Revenue by age group
- Sales by age group

Users can filter the dashboard by gender, category, subscription status, and shipping type.

This makes it easy to explore the data and understand customer behavior.

6. Business Recommendations

Based on the analysis, the following actions are suggested:

- Encourage more customers to subscribe
- Reward loyal customers with loyalty programs
- Focus marketing on customers who spend more
- Promote top-selling and highly rated products
- Use discounts carefully to protect profits

These actions can help the company increase sales and improve customer satisfaction.